

Evaluate the CBOW word embedding (3 parts)

Also read the additional write-up by Yue Lin

1. K-Nearest Neighbor evaluation

(a) Basic Idea

Well-trained word embeddings can capture the semantic properties of their corresponding words. If the semantic similarity between two words is strong, the cosine similarity between their embedding vectors should also be high. Therefore, we can evaluate word embeddings by finding k other word embeddings whose cosine similarities are the nearest to a query word and comparing whether the semantics between these k words and the query are similar.

(b) Practical Method

Select 50 commonly used words with a wide range of coverage as queries, find k other words with the nearest cosine similarity to the query, and compare their semantic similarities with the query. If most of the cases are similar, the evaluation results can be considered good. Also choose (randomly) 4 words and show their KNNs

In addition, on this basis, you can also average the k similarities selected for each query, and then average the average similarities of all queries. The higher the result, the better the word embedding effect is to a certain extent. However, this method may overly pursue higher similarity without a more rigorous standard similarity for comparison. This problem can be solved by the next evaluation method.

(c) Example

Query word: august
cosine similarity=0.8319: september
cosine similarity=0.8030: july
cosine similarity=0.7651: june

Figure 1: An example of the query and its 3 nearest similarities.

2. Word similarity golden standard evaluation

(a) Basic Idea

The cosine similarity between two words should not be as high as possible but adapt to the words' semantic similarity. For some word pairs, there is a certain degree of similarity between them but not very strong, so their cosine similarity should not be too low or too high. Therefore, the golden standard can be manually formulated to correctly measure the similarity between word pairs, and word embeddings can be evaluated by measuring how well the cosine similarity of word embeddings fits the golden standard.

(b) Practical Method

You randomly pick 20 pairs, find their embeddings and compute cosine, score=(cosine+1)x5

To provide a rigorous standard for the evaluation of word vectors, some researchers propose the word similarity golden standards. These golden standards list many word pairs (generally hundreds to thousands) and provide the semantic similarity of word pairs marked manually by experts as the standard similarity. When evaluating the embeddings, we can compare the cosine similarity pairs calculated using word embeddings with the standard similarity provided by the golden standard. The higher the correlation (generally represented by the Spearman Correlation Coefficient) between the standard and the calculated similarities, the better the word semantics are captured by embeddings, that is, the better the training effect should be.

big	heavy	6.18	Put here: $[\text{Cosine}(\text{embedding}(\text{big}), \text{embedding}(\text{heavy})) + 1] * 5$
rough	frigid	2.47	
bizarre	strange	9.37	
illegal	immoral	4.28	
bad	guilty	4.2	
modern	ancient	0.73	
new	ancient	0.23	
dull	funny	0.55	
happy	young	2	

Figure 2: A snapshot of the SimLex-999 golden standard.

(c) Example

Take the SimLex-999 golden standard as an example. There are 999 word pairs in SimLex-999 for each a standard similarity between 0 and 10 is provided. For the pair of “big” and “heavy”, the provided similarity is 6.18, showing that there is a certain similarity between the two, but not so significant. In contrast, the standard similarity between “big” and “large” is 9.55, which is visibly high. If there is a high correlation between the cosine similarity and the standard similarity generated by word embedding, then it can be considered that the word embedding “understands” the relative size of the semantic similarity between big and heavy and big and large. If this word embedding has a relatively thorough “understanding” of the semantic similarity of all word pairs selected by SimLex-999, then we can rigorously consider the performance of this word embedding to be good.

3. Analogy reasoning evaluation

You need to randomly pick 10 examples and show the results

(a) Basic Idea

Well-trained word embeddings can not only capture the semantic properties of a single word, but also preserve analogical relationships between words. From a more mathematical perspective, this means that if there are two pairs of words where the relationship within each pair is the same (like the relationship between a country and

its capital, or the relationship between the base form and the present participle of a word, etc.), then the result of subtraction between the embedding vectors of the first pair should be similar to that of the second pair. Therefore, word embeddings can be used to answer the question of “Given three words A, B, and C, find the word D such that the relationship between A and B is equivalent to the relationship between C and D”. This kind of question is called the analogical reasoning task, and its accuracy can serve as another criterion for evaluating word embeddings.

(b) Practical Method

For example, given words “man”, “woman”, and “king”, we have the following approximate equality

or

$$\text{vector}(\text{king}) - \text{vector}(\text{man}) = \text{vector}(\text{queen}) - \text{vector}(\text{woman})$$

normally this equation
do not hold exactly.
You show LHS and RHS

$$\text{vector}(\text{queen}) = \text{vector}(\text{king}) - \text{vector}(\text{man}) + \text{vector}(\text{woman})$$

Where $\text{vector}(*)$ means the embedding vector of the word “*”. Therefore, the answer given by the word embeddings to the example question should be the nearest embedding vector to

$$\text{vector}(\text{king}) - \text{vector}(\text{man}) + \text{vector}(\text{woman}) = \text{predicted vector}$$

where the distance is measured by cosine similarity. Note that the answer should exclude the given words, because the result calculated using these words is naturally similar to them, which will affect the evaluation results.

(c) Example

In practice, we usually use the analogical reasoning task dataset proposed by Mikolov in 2013. There are 8,869 semantic and 10,675 syntactic questions in the test set in total. The semantic questions can be divided into five categories and the syntactic questions can be divided into nine categories, allowing the dataset to be comprehensive of word relationships. The quality of word vectors is then measured by the total accuracy of all questions in this task.

The following lists the contents of each category of the aforementioned analogical reasoning task dataset for learning reference.

The semantic part contains the following categories:

- capital-common-countries
- capital-world
- currency
- city-in-state
- family

The syntactic part contains the following categories:

adjective-to-adverb
opposite
comparative
superlative
present-participle
nationality-adjective
past-tense
plural
plural-verbs